FOR A LOCALLY FINITE TYPE MORPHISM THE FLAT LOCUS IS OPEN AND THERE IT IS FLAT AND LOCALLY OF FINITE PRESENTATION, WHEN THE WEAKLY ASSOCIATED POINTS OF THE BASE ARE LOCALLY FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Locality of the question in X and S, so that the affine case suffices. Standard, used at proof line 4189. Flatness, openness of the flat locus and local finite presentation are local on source and target, so one may assume X and S affine.
- The algebra statement for a finite type ring map and a finite module over it. Same chapter flat.tex, label lemma-finite-type-flat-algebra (invoked at proof line 4193). Applied to B considered as a module over the polynomial algebra $A[x_1, \ldots, x_n]$ mapping onto it, it supplies openness of the flat locus and local finite presentation there.

Source reference. Original lemma 05IL: sources/stacks-flat/pinned-original-source.tex, source lines 4162–4162. Navigation pointer for \ref{lemma-finite-type-flat-algebra} used inside the counted span (line 4193); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Assume the set of weakly associated points of S is locally finite in S. Then the set of points $x \in X$ where f is flat is an open subscheme $U \subset X$ and $U \to S$ is flat and locally of finite presentation.*

Proof. The problem is local on X and S, hence we may assume that X and S are affine. Then $X \to S$ corresponds to a finite type ring map $A \to B$. Choose

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

a surjection $A[x_1, \ldots, x_n] \to B$ and consider B as an $A[x_1, \ldots, x_n]$-module. An application of Lemma 05IL finishes the proof. $\square$